

PALLADIUM BANK

Dimensional Model Design Written Explanation

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1. Introduction & Business Problem

Palladium Bank operates across Lagos, Abuja, Kano, Port Harcourt, and Ibadan. Currently, all reports run directly against raw 18-month transaction logs, causing slow performance and inconsistent metrics. This document outlines the design of a Star Schema dimensional model that enables the Head of Retail Banking to analyse fee income, channel behaviour, product performance, and customer churn with speed and consistency.

2. Data Profiling & Dimension Identification

Data profiling classified every column into its analytical role. The two core fact measures are Amount (NGN) — a fully additive measure — and Balance_After (NGN), which is semi-additive. The fact table grain is set at the individual transaction level (one row per Txn_ID), the most atomic level available. This supports all drill-down requirements and all four business objectives without pre-aggregation.

Five dimensions were identified: DIM_CUSTOMER (Customer_ID, Name, Tier), DIM_BRANCH (Branch_ID, Name, State, Region), DIM_PRODUCT (Product_ID, Name, Type), DIM_CHANNEL (Channel, Txn_Type), and DIM_DATE — a fully generated dimension covering all 547 days in the 18-month window. Txn_ID is retained as a degenerate dimension directly in the fact table, serving as a unique audit reference without requiring its own table.

3. Slowly Changing Dimensions (SCD)

DIM_CUSTOMER — SCD Type 2

Customer Tier is implemented as SCD Type 2 because it changes over time and is business-critical. When a customer moves from Silver to Gold, the old record is expired and a new one is inserted. This prevents historical rewriting — every transaction can be accurately joined to the tier held at that point in time, which is essential for answering whether high-value customers were reducing activity before a tier change.

DIM_BRANCH — SCD Type 1

Branch attributes are managed with SCD Type 1. Branch name and region updates are rare administrative corrections with no analytical significance. No business question requires knowing a branch's previous name, so a simple overwrite is sufficient and keeps the model lean.

4. ETL/ELT Strategy & Data Quality

The initial load processes 18 months of data in strict dependency order: DIM_DATE first, then the static lookup dimensions, then DIM_CUSTOMER, and finally FACT_TRANSACTIONS. Key transformations include converting Txn_Date to a YYYYMMDD date_key, stripping commas from Amount, extracting txn_hour, and deriving Region from State.

Incremental daily loads use Txn_ID as a deduplication key to detect new records. SCD Type 2 updates are handled by closing the current flag on old customer records and inserting new active rows. Four data quality checks are enforced: referential integrity (orphan Customer_IDs are quarantined), null validation (missing Amount or Branch_ID rows are rejected), signage consistency (negative deposit amounts are flagged), and duplicate detection (repeat Txn_IDs are dropped).

5. Hierarchies, Performance & Scalability

Three drill-down hierarchies support the required analytical views: Time (Year to Quarter to Month to Day), Geography (State to Branch), and Product (Product_Type to Product_Name). These are supported by DIM_DATE, DIM_BRANCH, and DIM_PRODUCT respectively.

FACT_TRANSACTIONS is partitioned by month on date_key so queries scan only relevant partitions, improving MoM and QoQ report performance as data grows. Six indexes on foreign key columns accelerate all dimension joins. A pre-aggregated table, AGG_MONTHLY_BRANCH_REVENUE, pre-calculates total transactions, total amount, and average transaction value per branch per month — powering executive dashboards without scanning the full fact table on every run.

6. Conclusion

This Star Schema design transforms Palladium Bank's raw transaction logs into a structured analytics platform. By applying SCD Type 2 for customer history, enforcing data quality checks, and implementing partitioning and aggregation strategies, the bank can now answer dynamic questions around customer behaviour, product performance, and churn risk with the speed and accuracy that executive decision-making demands.